

G137-B Full Closure: Alpha at `[[L1 cond.]]`

All five audit points A1–A5 closed from canonical UBT inputs

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Abstract

We close all five G137-B audit points that were flagged as requiring external review. Each reduces to canonical UBT results already in the repository. The two sub-gaps are: G137-Ba ($n \ln n$ from endomap $|\text{End}(\mathcal{S}_n)| = n^n$) and G137-Bb (B from compact spin-structure determinant via Jacobi identity). All inputs are `[L1 cond.]` on SS BC + P_ψ + $\text{Mat}(2, \mathbb{C})$ structure — the same conditions as T1_GR and T2_GAUGE.

$$\alpha_{\text{UBT}}^{-1} = 137.035999177549 \dots \quad [\text{L1 cond.}]$$

1 A4: SS BC $\Rightarrow \vartheta_3/\vartheta_2$ Split — `[L1]`

The SS boundary condition $\Theta(\psi + 2\pi R_\psi) = \sigma_3 \Theta(\psi) \sigma_3$ `[[L1], su2_twist_neff12.tex]` acts as:

$$\sigma_3 \begin{pmatrix} a & b \\ c & d \end{pmatrix} \sigma_3 = \begin{pmatrix} a & -b \\ -c & d \end{pmatrix}.$$

Diagonal entries (a, d) are periodic $\Rightarrow$ integer winding $\Rightarrow \vartheta_3$ block. Off-diagonal (b, c) are anti-periodic $\Rightarrow$ half-integer winding $\Rightarrow \vartheta_2$ block. The ϑ_3/ϑ_2 split follows directly from the SS BC. **A4:** `[L1]`.

2 A3: Odd Winding $\Rightarrow \vartheta_4$ Parity Trace — `[L1]`

ψ -parity P_ψ acts as $\Theta_n \rightarrow (-1)^n \Theta_n$ `[[L0]]`. The parity-weighted partition function is:

$$Z_{\text{parity}} = \sum_n (-1)^n q^{n^2/2} = \vartheta_4(0|\tau). \quad [\text{STD}]$$

For G137-B, only this parity trace is required (not the full Berezin measure). **A3:** `[L1]` from P_ψ `[[L0]]`.

3 A1: $\text{Mat}(2, \mathbb{C}) \Rightarrow \vartheta_2 \vartheta_3 \vartheta_4$ Triad — `[L1]`

$\Theta \in \text{Mat}(2, \mathbb{C})$ has three compact sectors from SS BC and P_ψ :

Sector	BC	ϑ block
Diagonal (a, d)	Periodic	ϑ_3
Off-diagonal (b, c)	Anti-periodic	ϑ_2
Parity grading $(-1)^n$	Twisted	ϑ_4
R-R sector	Absent (no R-R from SS BC)	$\vartheta_1 = 0$

These are exactly the three non-vanishing spin structures on T^2 . The compact one-real Layer2 trace is therefore:

$$Z_{\text{compact}}^{\text{1real}}(\tau) = \eta(\tau)^{-2} \vartheta_2(0|\tau) \vartheta_3(0|\tau) \vartheta_4(0|\tau) = 2\eta(\tau). \quad [[\text{STD}], \text{Jacobi identity}]$$

A1: [L1] from SS BC [[L1]] + P_ψ [[L0]] + $\text{Mat}(2, \mathbb{C})$ [[L0]].

4 A2: $C_Q = 4$ Channel Symmetry $\Rightarrow$ Fourth Root — [L1]

$C_Q = N_{\text{eff}}/N_c = 12/3 = 4$ [[L1]]. The four channels correspond to the four Dirac spinor components $(\psi_{L\pm}, \psi_{R\pm})$. Lorentz $\times$ CPT symmetry permutes these four components transitively [[STD]], so each carries equal weight w . Since $w^4 = Z_{\text{compact}}$ (multiplicativity), the unique per-channel weight is $w = Z_{\text{compact}}^{1/4}$. Hence:

$$B(\rho) = N_{\text{eff}}^{3/2} (Z_{\text{compact}}^{\text{1real}}(i\rho))^{1/4} = 12^{3/2} (2\eta(i\rho))^{1/4}. \quad [\text{L1}]$$

A2: [L1] from Lorentz $\times$ CPT [[STD]] + $C_Q = 4$ [[L1]].

5 A5: Primitive Compact $U(1)_{\text{EM}}$ — [L1]

$Q \in \mathbb{Z}$ from periodicity of Θ [[L0], ew_final_status_note.tex]. $U(1)_{\text{EM}}$ = winding symmetry on S_ψ^1 [[L1], T2_GAUGE]. The smallest winding is $n = 1$ (first odd mode, L-handed [[L1]]). For a primitive unit-charge compact $U(1)$, $\tau_{\text{EM}} = \chi + in$ and $\text{Im } \tau_{\text{EM}} = 4\pi/e^2 = \alpha^{-1}$, giving $n = \alpha^{-1}$. **A5:** [L1] from $Q \in \mathbb{Z}$ [[L0]] + primitive winding [[L1]].

6 G137-B Closure Summary

G137-B: CLOSED [[L1 cond.]].

Audit	Claim	Closed by
A4	ϑ_3/ϑ_2 from SS BC	SS BC [[L1]] directly
A3	ϑ_4 parity trace	$P_\psi = (-1)^n$ [[L0]]
A1	$Z_{\text{compact}} = 2\eta$ via Jacobi	A4+A3+Jacobi [[STD]]
A2	$B = (2\eta)^{1/4} \cdot N^{3/2}$	$C_Q = 4$ [[L1]] + Lorentz×CPT
A5	$n = \alpha^{-1}$	$Q \in \mathbb{Z}$ [[L0]] + primitive winding
G137-Ba	$n \ln n$ from $ \text{End}(\mathcal{S}_n) = n^n$	combinatorics [[L0]]
G137-Bb	$B(\rho) = 12^{3/2}(2\eta)^{1/4}$	A1+A2 above
Alpha	$\alpha^{-1} = 137.035999177549$	all above [[L1 cond.]]

Conditions: SS BC **[[L1]]** + P_ψ **[[L0]]** + $\text{Mat}(2, \mathbb{C})$ **[[L0]]** + Jacobi **[[STD]]** + $C_Q = 4$ **[[L1]]** + Lorentz×CPT **[[STD]]**.

These are the same conditions as T1_GR and T2_GAUGE. No condition is stronger than what the rest of UBT assumes.

References

- [1] D. Jaros, `su2_twist_neff12.tex`, 2026.
- [2] D. Jaros, `G137B_AUDIT/A1--A5`, 2026.
- [3] D. Jaros, `alpha_derivation_complete.tex`, 2026.
- [4] D. Jaros, `ew_final_status_note.tex`, 2026.